

STILLWATER 2 W, OK, USA

WMO: 746420

Lat: 36.1181N

Lon: 97.0914W

Elev: 271

StdP: 98.11

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53926

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.8	-8.2	-17.2	0.9	-9.2	-14.6	1.1	-6.3	8.5	3.8	7.6	4.2	2.5	N/A	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	38.3	23.3	36.4	23.4	34.9	23.5	25.6	33.5	25.1	32.8	24.5	32.1	2.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	18.9	29.4	22.9	18.2	28.9	22.3	17.6	28.4	80.3	33.4	77.8	32.9	75.4	32.1	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.2	6.2	5.5	DB	-15.7	40.0	3.9	2.0	-18.5	41.4	-20.8	42.6	-23.0	43.7	-25.8	45.2	(4)
(5)			WB	-17.1	26.5	4.7	0.8	-20.5	27.1	-23.2	27.6	-25.9	28.0	-29.3	28.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.9	3.3	4.8	11.1	15.7	20.5	25.7	28.0	27.4	23.1	16.2	9.9	4.3	(6)	
	DBStd	10.03	5.83	6.38	5.99	5.12	4.48	3.01	3.05	3.23	3.92	5.22	5.26	5.41	(7)	
	HDD10.0	716	219	166	57	11	0	0	0	0	0	9	64	187	(8)	
	HDD18.3	2024	467	379	234	109	31	1	0	0	7	105	257	434	(9)	
	CDD10.0	2867	11	21	92	182	326	472	558	540	394	201	61	12	(10)	
	CDD18.3	1134	0	1	10	31	97	222	299	282	151	37	3	0	(11)	
	CDH23.3	12968	2	12	90	318	946	2489	3758	3363	1572	381	36	1	(12)	
(13)	CDH26.7	6261	0	2	16	83	328	1158	2065	1806	689	111	4	0	(13)	
(14)	Wind	WSAvg	2.3	2.4	2.7	2.8	2.9	2.4	1.9	1.9	1.9	2.1	2.2	2.3	(14)	
(15) Precipitation	PrecAvg	879	31	36	69	78	124	109	76	78	103	76	58	40	(15)	
	PrecMax	1179	93	137	196	211	371	268	209	216	315	222	169	130	(16)	
	PrecMin	500	0	2	0	4	15	21	0	1	1	10	0	2	(17)	
	PrecStd	171	27	31	47	49	72	60	49	54	66	53	47	31	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.2	25.1	28.8	31.8	33.9	37.7	40.7	41.2	36.9	31.8	26.9	21.6	(19)	
		MCWB	11.9	14.5	17.2	19.3	21.5	23.8	22.6	22.7	22.0	21.4	16.7	13.4	(20)	
	2%	DB	19.5	21.8	25.8	29.1	32.0	35.3	38.6	38.2	34.1	29.7	24.0	18.7	(21)	
		MCWB	11.2	13.5	15.7	17.7	21.4	23.6	22.9	23.7	22.4	19.3	15.3	12.8	(22)	
	5%	DB	16.6	18.8	23.5	26.9	30.3	33.8	36.6	36.4	32.6	27.6	21.7	15.8	(23)	
		MCWB	10.0	12.6	14.6	16.8	20.8	23.6	23.5	23.7	22.4	18.4	14.4	10.0	(24)	
	10%	DB	13.4	15.7	21.1	24.9	28.4	32.4	35.0	34.5	30.7	25.3	19.2	13.7	(25)	
		MCWB	7.5	10.1	13.6	16.9	20.3	23.2	23.8	23.2	21.6	17.7	12.9	8.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.9	16.9	19.5	21.1	24.3	26.0	26.7	26.1	24.8	22.7	19.4	16.8	(27)	
		MCDB	18.0	21.3	26.1	27.1	30.6	34.2	33.8	34.7	32.1	29.3	23.4	20.2	(28)	
	2%	WB	11.6	15.4	17.9	19.9	23.1	25.4	25.7	25.4	23.9	21.7	16.8	13.5	(29)	
		MCDB	17.0	20.4	24.0	25.8	29.4	32.9	33.1	34.1	31.5	27.3	21.2	15.6	(30)	
	5%	WB	9.6	13.4	16.5	19.0	22.2	24.8	25.1	24.8	23.3	20.7	15.4	11.2	(31)	
		MCDB	15.7	18.1	22.0	24.9	28.2	32.0	32.6	33.0	30.6	25.5	20.0	14.9	(32)	
	10%	WB	7.7	10.4	14.5	17.9	21.4	24.1	24.5	24.1	22.7	19.3	13.8	8.8	(33)	
		MCDB	12.4	16.0	19.7	23.4	26.9	31.0	32.1	31.8	29.4	23.7	17.8	12.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.7	12.4	13.0	12.7	11.5	11.0	11.6	11.9	12.4	12.9	13.1	11.8	(35)	
		MCDBR	18.4	17.2	15.7	15.4	13.2	12.6	13.9	14.4	13.8	15.1	16.2	15.9	(36)	
	5% WB	MCWBR	10.9	10.4	8.5	7.8	5.5	4.2	3.7	4.1	4.8	6.7	8.9	10.2	(37)	
		MCWBR	16.7	15.2	13.1	12.7	10.9	10.9	11.2	11.8	11.2	11.3	12.3	12.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.297	0.302	0.342	0.382	0.419	0.429	0.434	0.436	0.393	0.348	0.314	0.291	(40)	
		taud	2.496	2.494	2.411	2.315	2.284	2.291	2.316	2.254	2.358	2.470	2.504	2.524	(41)	
	Edn at Noon		906	942	927	901	868	855	848	839	861	876	878	891	(42)	
		Edh at Noon	81	92	109	127	133	132	128	134	114	93	80	75	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.68	3.31	4.32	5.37	5.89	6.73	6.74	6.10	5.07	3.82	2.84	2.26	(44)	
	RadStd		0.27	0.49	0.28	0.37	0.59	0.60	0.37	0.37	0.42	0.47	0.36	0.24	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page